Exact Cycle Classification for Every Finite Kac Ring

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Abstract

For every ring size and every binary marker configuration, we classify all cycles of the reversible finite Kac ring. The marker product η is the only invariant needed: positive product gives two cycles of length N, while negative product gives one cycle of length $2N$. We derive all fixed counts, the Artin–Mazur zeta, and the finite Koopman determinant.

Keywords: Kac ring; marker configuration; exact period; primitive cycle; dynamical zeta; Koopman permutation; time reversal.

中文摘要

本文对任意环长和任意二元标记配置的可逆有限Kac环给出完整周期分类。 所有动力信息只依赖标记乘积 η：正乘积产生两个长度为 N 的周期，负乘积产生 一个长度为 $2N$ 的周期。由此严格导出所有时刻的不动点数、动力 ζ 函数、 有限维Koopman行列式与反酉时间反演。

关键词：Kac环；标记配置；精确周期；本原周期；动力 ζ 函数； Koopman置换；时间反演。

1 All-marker classification

On $X_N = \mathbb{Z}/N\mathbb{Z} \times \{\pm 1\}$ let

$$T(j, s) = (j + 1, \varepsilon_j s), \qquad \eta = \prod_{j=0}^{N-1} \varepsilon_j.$$

One circuit gives $T^N(j, s) = (j, \eta s)$. Since every return time is divisible by N, all states have exact period N when $\eta = +1$ and exact period $2N$ when $\eta = -1$. Hence there are respectively two and one geometric cycles. With $L = N$ or $2N$ and $c = 2N/L$,

$$\# \operatorname{Fix}(T^n) = 2N\, \mathbf{1}_{L|n}, \quad \zeta_T(z) = (1 - z^L)^{-c}, \quad \det(I - zU_T) = (1 - z^L)^c.$$

The eigenvalues of the counting-measure Koopman permutation are all L-th roots of unity, each with multiplicity c.

2 Gauge and reversal

Set $g_0 = 1$, $g_j = \prod_{r<j} \varepsilon_r$, and $q = g_j s$. In (j, q) coordinates the map advances the site, preserves q away from the wrap, and sends q to ηq at the wrap. If $\eta = +1$, reflection $j \mapsto -j$ reverses each cycle. If $\eta = -1$, define $\psi(j, +1) = j$ and $\psi(j, -1) = N + j$; the map becomes translation by one on $\mathbb{Z}/(2N)\mathbb{Z}$, reversed by $t \mapsto -t$. Pullback through the gauge yields an involution R with $RTR = T^{-1}$. Thus $\Theta f = \overline{f \circ R}$ is antiunitary and $\Theta U_T \Theta = U_T^{-1}$. The permutation unitary is self-adjoint exactly when $L \leq 2$.

3 Boundary

The classification includes $N = 1$. It is an exact finite kinetic toy model, not a target divisor or Hilbert–Pólya construction. Route B is not authorized. Scope: `NO_BAD_EULER_OR_ROOT_NUMBER`.